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8 **(a)** By considering the binomial expansion of $\left(z+\frac{1}{z}\right)^7$, where $z=\cos\theta+\mathrm{i}\sin\theta$, use de Moivre's theorem to show that

$$\cos^7\theta = a\cos 7\theta + b\cos 5\theta + c\cos 3\theta + d\cos\theta,$$

where a, b, c and d are constants to be determined. [5]